Iterative Improvement: Stable Matching

Design and Analysis of Algorithms

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Iterative improvement

- *Iterative improvement technique* starts with some feasible solution (a solution that satisfies all the constraints of the problem).
- It then proceeds to improve the feasible solution by repeated applications of some simple step.
- This step typically involves a small, localized change yielding a feasible solution with an improved value of the objective function.
- When no such change improves the value of the objective function, the algorithm returns the last feasible solution as optimal and stops.

Challenges

- We need an initial feasible solution.
- It is not always clear what changes should be allowed in a feasible solution so that we can check efficiently whether the current solution is locally optimal and, if not, replace it with a better one.
- There is the issue of local versus global extremum (maximum or minimum).
- Yet, there are important problems that can be solved by iterative improvement algorithms like linear programming.

Stable matching / Stable marriage problem

- Consider a set $Y = \{m_1, m_2, \ldots, m_n\}$ of n men and a set $X = \{w_1, w_2, \ldots, w_n\}$ of n women. Each man has a preference list ordering the women as potential marriage partners with no ties allowed. Similarly, each woman has a preference list of the men, also with no ties.
- A marriage matching M is a set of $n(m, w)$ pairs whose members are selected from disjoint n-element sets Y and X in a one-one fashion, i.e., each man m from Y is paired with exactly one woman w from X and vice versa.

Representing preferences

- The preferences can be presented by an $n \times n$ ranking matrix.

men's preferences

	1st	2nd	3rd
Bob:	Lea	Ann	Sue
Jim:	Lea	Sue	Ann
Tom:	Sue	Lea	Ann

(a)

women's preferences

	1st	2nd	3rd
Ann:	Jim	Tom	Bob
Lea:	Tom	Bob	Jim
Sue:	Jim	Tom	Bob

(b)

ranking matrix

	Ann	Lea	Sue
Bob	2,3	1,2	3,3
Jim	3,1	1,3	2,1
Tom	3,2	2,1	1,2

(c)

- The rows and columns of the matrix represent the men and women of the two sets, respectively.
- A cell in row m and column w contains two rankings: the first is the ranking of w in the m's preference list; the second is the ranking of m in the w's preference list.
- For example, the pair 3, 1 in Jim's row and Ann's column in the matrix indicates that Ann is Jim's third choice while Jim is Ann's first.

Blocking pair

- A pair (m, w) is said to be a blocking pair for a marriage matching M if man m and woman w are not matched in M but they prefer each other to their mates in M.
- A marriage matching M is called *stable* if there is no blocking pair for it. That is, there are no two people of opposite gender who would both rather marry each other than their current partners allotted by the matching.

Example

men's preferences

	1st	2nd	3rd
Bob:	Lea	Ann	Sue
Jim:	Lea	Sue	Ann
Tom:	Sue	Lea	Ann

(a)

women's preferences

	1st	2nd	3rd
Ann:	Jim	Tom	Bob
Lea:	Tom	Bob	Jim
Sue:	Jim	Tom	Bob

(b)

ranking matrix

	Ann	Lea	Sue
Bob	$\boxed{2,3}$	1,2	3,3
Jim	3,1	$\boxed{1,3}$	2,1
Tom	3,2	2,1	$\boxed{1,2}$

(c)

Example

men's preferences

	1st	2nd	3rd
Bob:	Lea	Ann	Sue
Jim:	Lea	Sue	Ann
Tom:	Sue	Lea	Ann

(a)

women's preferences

	1st	2nd	3rd
Ann:	Jim	Tom	Bob
Lea:	Tom	Bob	Jim
Sue:	Jim	Tom	Bob

(b)

ranking matrix

	Ann	Lea	Sue
Bob	$\boxed{2,3}$	1,2	3,3
Jim	3,1	$\boxed{1,3}$	2,1
Tom	3,2	2,1	$\boxed{1,2}$

(c)

- Consider the matching M = {(Bob, Ann), (Jim, Lea), (Tom, Sue)}
- (Bob, Lea) is a blocking pair for M because they are not matched in M while Bob prefers Lea to Ann and Lea prefers Bob to Jim.
- M is **unstable** because Bob and Lea can drop their designated mates to join in a union they both prefer.
- The stable marriage problem is to find a stable marriage matching for men's and women's given preferences.
- Surprisingly, this problem always has a solution.

Tracing the example

Free men:
Bob, Jim, Tom

	Ann	Lea	Sue
Bob	2, 3	$\boxed{1,2}$	3, 3
Jim	3, 1	1, 3	2, 1
Tom	3, 2	2, 1	1, 2

Bob proposed to Lea
Lea accepted

Free men:
Jim, Tom

	Ann	Lea	Sue
Bob	2, 3	$\boxed{1,2}$	3, 3
Jim	3, 1	1, 3	2, 1
Tom	3, 2	2, 1	1, 2

Jim proposed to Lea
Lea rejected

Free men:
Jim, Tom

	Ann	Lea	Sue
Bob	2, 3	$\boxed{1,2}$	3, 3
Jim	3, 1	1, 3	$\boxed{2,1}$
Tom	3, 2	2, 1	1, 2

Jim proposed to Sue
Sue accepted

Free men:
Tom

	Ann	Lea	Sue
Bob	2, 3	$\boxed{1,2}$	3, 3
Jim	3, 1	1, 3	$\boxed{2,1}$
Tom	3, 2	2, 1	1, 2

Tom proposed to Sue
Sue rejected

Free men:
Tom

	Ann	Lea	Sue
Bob	2, 3	1, 2	3, 3
Jim	3, 1	1, 3	$\boxed{2,1}$
Tom	3, 2	$\boxed{2,1}$	1, 2

Tom proposed to Lea
Lea replaced Bob with Tom

Free men:
Bob

	Ann	Lea	Sue
Bob	$\boxed{2,3}$	1, 2	3, 3
Jim	3, 1	1, 3	$\boxed{2,1}$
Tom	3, 2	$\boxed{2,1}$	1, 2

Bob proposed to Ann
Ann accepted

Finally, the algorithm

Input: A set of n men and a set of n women along with rankings of the women by each man and rankings of the men by each woman with no ties allowed in the rankings

Output: A stable marriage matching

Stable marriage algorithm

Step 0 Start with all the men and women being free.

Step 1 While there are free men, arbitrarily select one of them and do the following:

Proposal The selected free man m proposes to w, the next woman on his preference list (who is the highest-ranked woman who has not rejected him before).

Response If w is free, she accepts the proposal to be matched with m. If she is not free, she compares m with her current mate. If she prefers m to him, she accepts m's proposal, making her former mate free; otherwise, she simply rejects m's proposal, leaving m free.

Step 2 Return the set of n matched pairs.

Exercise

Find a stable matching for this set of preferences, where $Y = \{A, B, C, D, E\}$ is the set of men, and $X = \{L, M, N, O, P\}$ is the set of women.

A	O	M	N	L	P
B	P	N	M	L	O
C	M	P	L	O	N
D	P	M	O	N	L
E	O	L	M	N	P

L	D	B	E	C	A
M	B	A	D	C	E
N	A	C	E	D	B
O	D	A	C	B	E
P	B	E	A	C	D

Time complexity

- The stable marriage algorithm terminates after no more than $O(n^2)$ iterations with a stable marriage output.
  Why?

Summary

- The stable marriage problem is to find a stable marriage matching for men's and women's given preferences.
- A marriage matching M is called stable if there is no blocking pair for it; otherwise, M is called unstable.
- The stable marriage algorithm (in this lecture) iterates through all free men until any free men are available.
- The selected free man m proposes to w, the next woman on his preference list.
- If w is free, she accepts the proposal to be matched with m. If she is not free, she compares m with her current mate. If she prefers m to him, she accepts m's proposal, making her former mate free; otherwise, she simply rejects m's proposal, leaving m free.
- No man proposes to a woman more than once.
- Stability is a matching property that can be desirable in a variety of applications.